

Titanic Dataset - EDA Summary

Overall Survival Rate

The number of people who did not survive is significantly higher than those who did. Only about 38% survived, indicating the disaster's severity.

Survival by Gender

Females had a much higher survival rate than males. This suggests a likely 'women and children first' rescue policy.

Survival by Passenger Class

1st Class passengers had the highest survival rate, followed by 2nd, then 3rd. Passengers in 3rd class were least likely to survive, suggesting socio-economic bias in rescue efforts.

Age Distribution

Most passengers were between 20 and 40 years old. The distribution is right-skewed with fewer older passengers.

Age vs Survival

Children (especially under 10 years) had relatively better chances of survival. Younger females especially seem to have been prioritized.

Fare Distribution

Most fares were on the lower end, but some high outliers exist (luxury passengers). Survivors generally had a slightly higher fare distribution—again correlating with class.

Missing Values

Notably missing values include: Cabin (many NaNs), Age (some missing), and Embarked (very few missing).

Correlation Matrix

Fare and Pclass are negatively correlated. Survived has a weak positive correlation with Fare and Pclass. Very little correlation between Age and survival.

Survival by Embarkation Port

Most passengers embarked from Southampton (S). Highest survival rates were among passengers from Cherbourg (C), likely because more 1st class passengers boarded there.

Pairplot Patterns

Reinforces trends in age, fare, class, and gender differences with respect to survival. Shows clusters of survivors based on combinations of features.